DEMO: USING THE PROJECT SHARES

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GHENT UNIVERSITY Data Stewards

CC BY

TYPE OF PLATFORMS

Local Storage

- Laptop
- Desktop
- USB drive
- External HD





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Network Storage

- Data on UGent-servers



Cloud Storage

- ….in the cloud…

OneDrive for Business

Dropbox

TYPE OF PLATFORMS

Local Storage

Advantages:

- Easy/direct access
- Immediately at hand
- Off-line use

Disadvantage:

- Prone to loss
- Hardware failure
- You have to arrange everything yourself

Network Storage

Advantages:

- DICT is in charge of:
  - Security/maintenance
  - Backup
- Safe for confidential data
- Sustainable accces

Disadvantage:

- You need to be on the UGent network
  - Either local
  - Or through VPN

Cloud Storage

Advantages:

- Accessible anywhere
- Usually automatic syncing
- Off-line use

Disadvantage:

- Confidentiality?
- Safety?
- Sustainable?

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CENTRAL DISK SPACE – WHAT DICT OFFERS

See https://helpdesk.ugent.be/netdisk/

- Personal
  - personal disk space: H:-Drive (quota: 5 GB)
  - personal web space: W:-Drive (quota: 1 GB)

- Shares
  - Shared network drive: S:-drive (quota: on demand, GB-TB range)

    Intended for project-related storage of data

    e.g. ‘vakgroep’-shares

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	Office 365			Shares		HPC	
Name	OneDrive FB	SharePoint	Teams SharePoint	H:/	Project Share (acl & bulk)	HOME/DATA/SCRATCH	VO_DATA/VO_SCRATCH
Location	Cloud	UGent	Cloud	UGent	UGent	UGent	UGent
Size	2TB	5GB	See link	5GB	GB-TB range	25 GB	GB-TB range
Backup mechanism	Versioning	Versioning	Versioning	snapshots	snapshots	none	none
Sync to local	✓		✓				
Confidential	Encrypt	✓	Encrypt	✓	✓	✓	✓
Intended use	Personal	Sharing	Sharing	Personal	Sharing	Personal	Sharing
Sharing external users?	✓	✓	✓				✓
Suitable for TB datasets					✓		✓
Data in use	✓	✓	✓	✓	✓	✓	✓
Data at rest					✓		

REQUESTING A SHARE

https://helpdesk.ugent.be/netdisk/en/shares.php

UGent staff can request additional shared network drives:

- intended for project-related storage of data (e.g. for research purposes)
- not meant for distributing webpages or public files
- will be provided in the form of a share (without url)
- DICT is in charge of the set-up, maintenance and the back-up of the server
- on the shares "snapshots" of the data are made. Snapshots allow you to retrieve (older versions of) files, without having to contact DICT. To retrieve previous versions you can go back 15 weeks in total, based on
  - 5 weeks of daily snapshots
  - 15 weeks of weekly snaphots
- employees can request additional shared network drives using the webform.
- Access to different users
- to increase disk quota (the amount of disk space): send an e-mail to helpdesk@UGent.be

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Aanvraagformulieren

Vakgroepshare

Nieuwe vakgroepshare aanvragen

UGent medewerkers kunnen gedeelde netwerkschijven ("shares") bekomen:
→ voor "projectgebonden" opslag van data (bv. voor onderzoeksdoeleinden)

→ **niet** voor het aanbieden van webpagina's of publieke files

→ DICT staat in voor het opzetten en onderhoud van de server en voorziet back-ups

→ op de shares worden "snapshots" van de data gemaakt. Die zorgen ervoor dat u zelf in staat bent om (oudere versies van) bestanden terug te zoeken.

Share gegevens:

Geef de naam op die u wenst voor de share. *

Geef een beknopte omschrijving van de share. Geef ook op waar de eventuele afkorting voor staat. *

Gewenste grootte: 50 GB

Gebruikers:

Wie moet toegang krijgen tot de netwerkschijf of share?
Enkel UGent-medewerkers komen hiervoor in aanmerking.
Het opgeven van een verantwoordelijke is verplicht.

Verantwoordelijke: * tkvdevel

Gebruikers die toegang moeten hebben tot de share (optioneel):

Gebruikersnaam, e-mail adres of naam: Zoek UGent gebruiker.. +

Opmerkingen:

* verplicht in te vullen

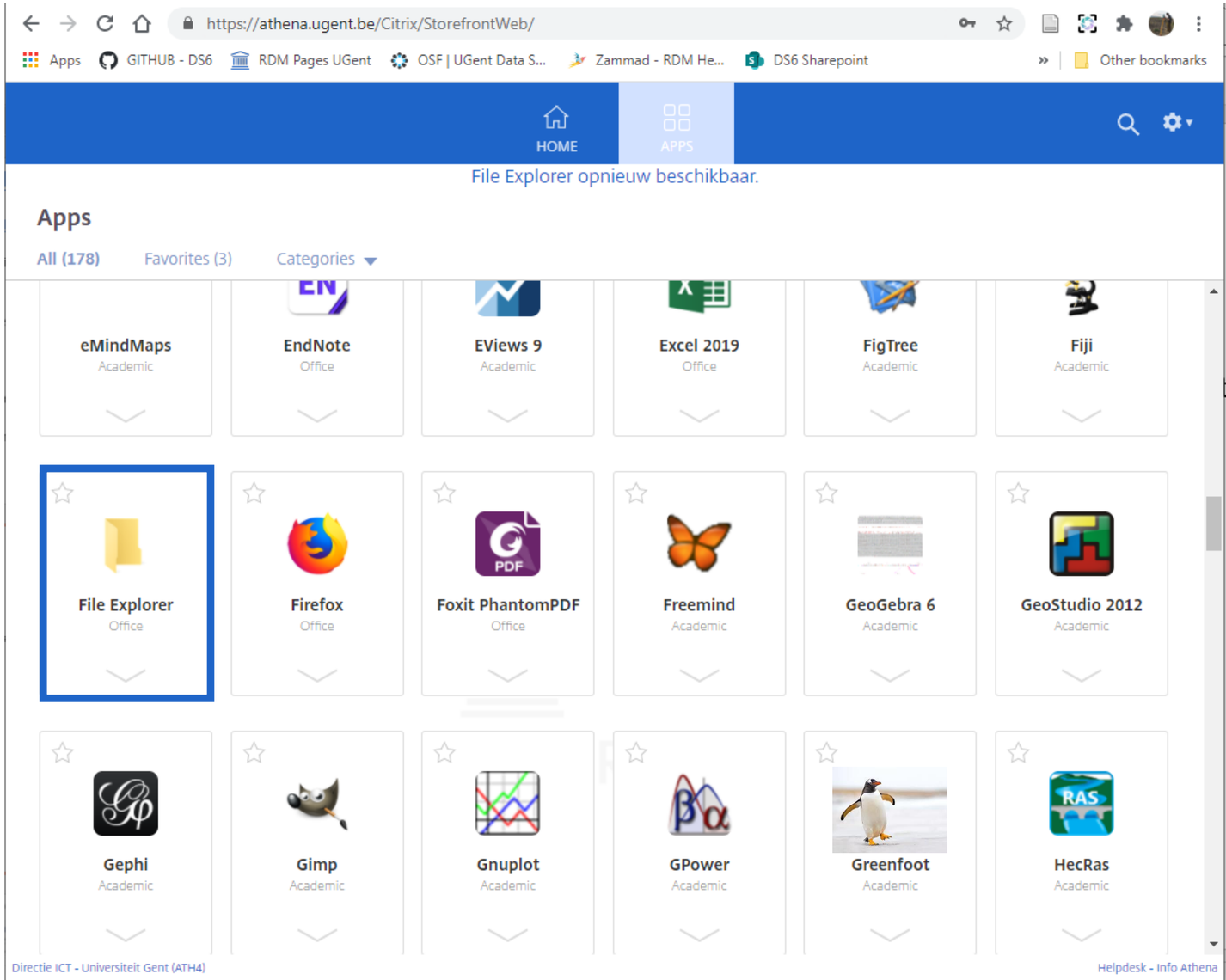
Verzend uw aanvraag

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MOUNTING SHARES

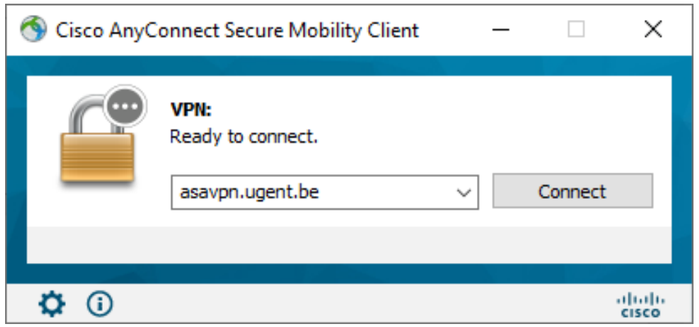
https://helpdesk.ugent.be/netdisk/bestand_mount.php

- Via Athena
- Mounting on your local computer
  - Requires VPN

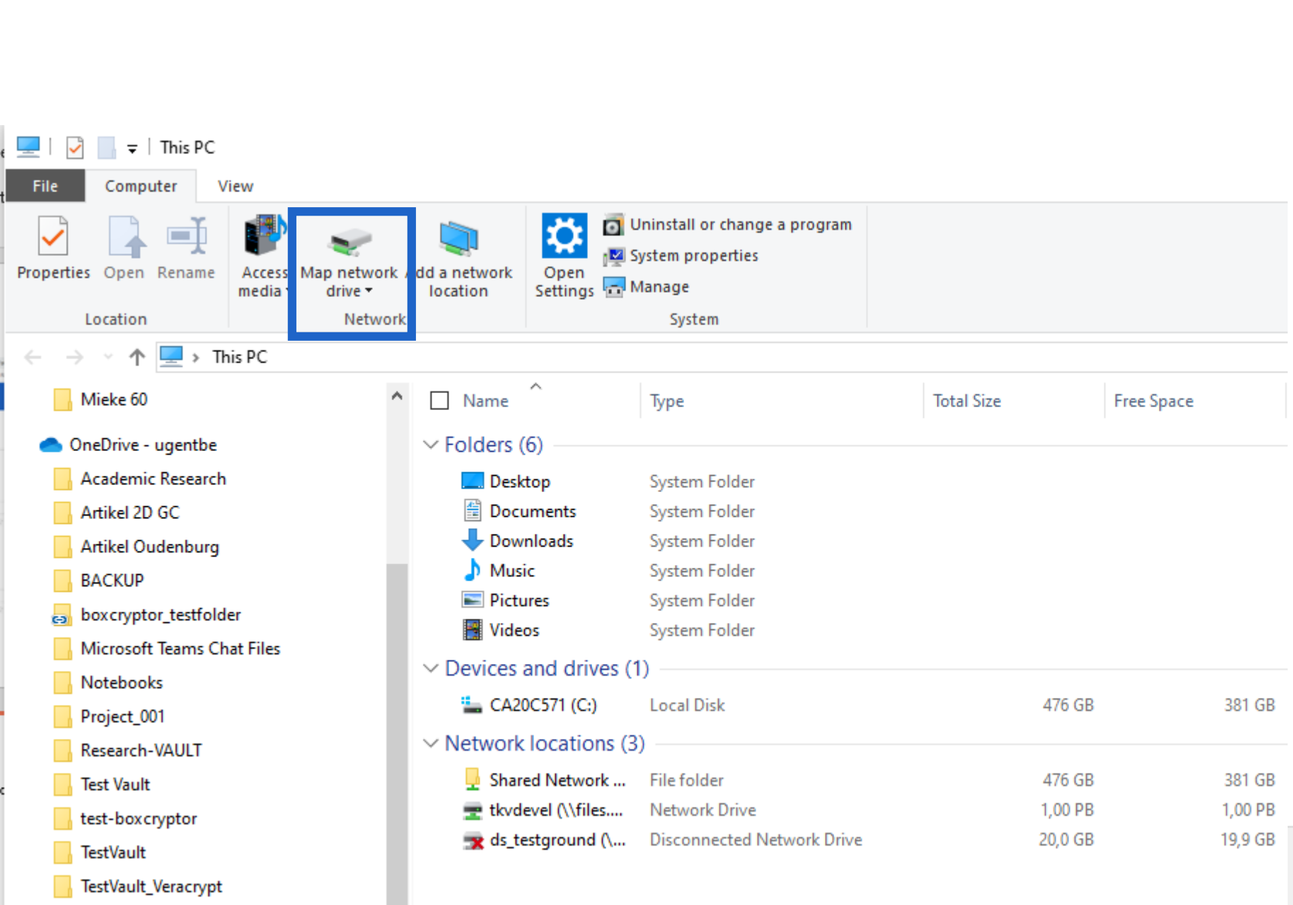


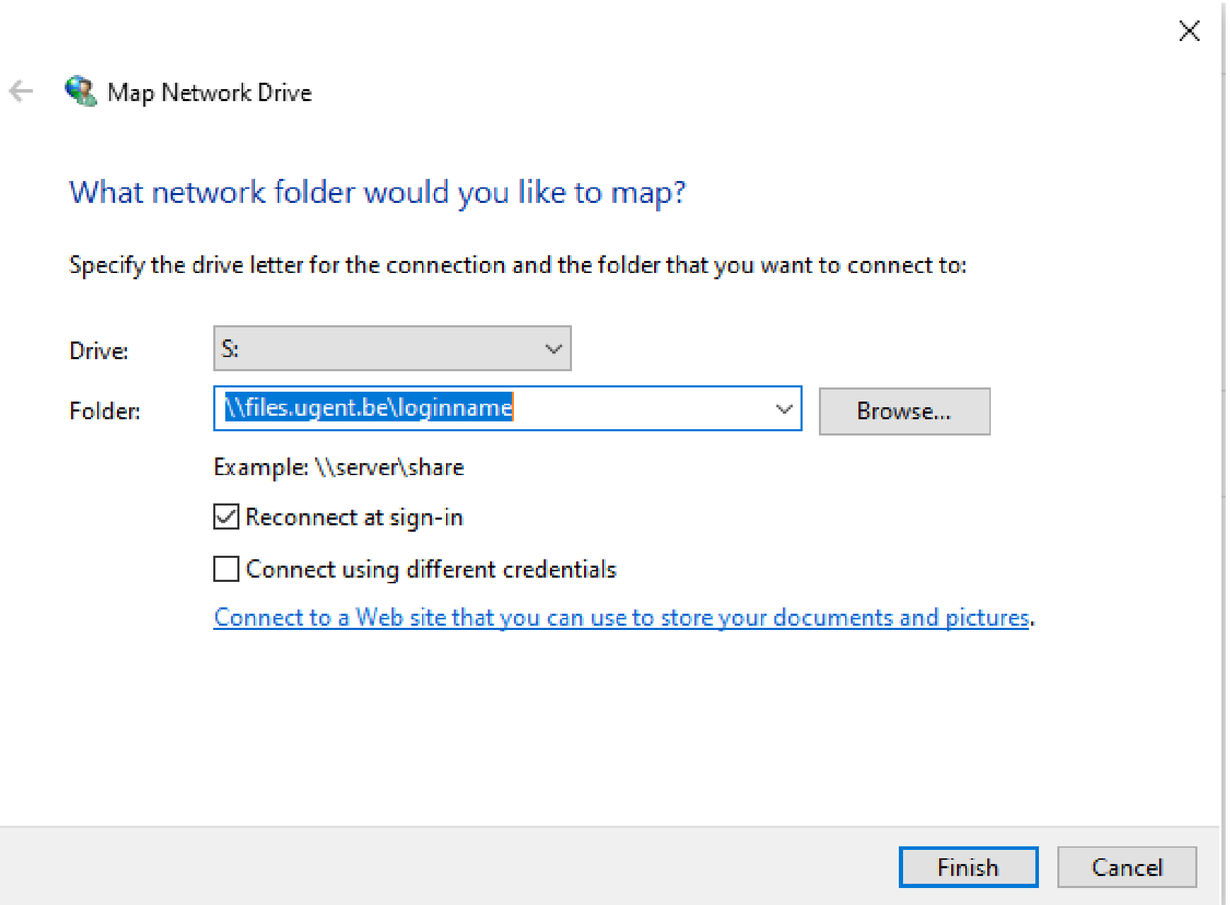
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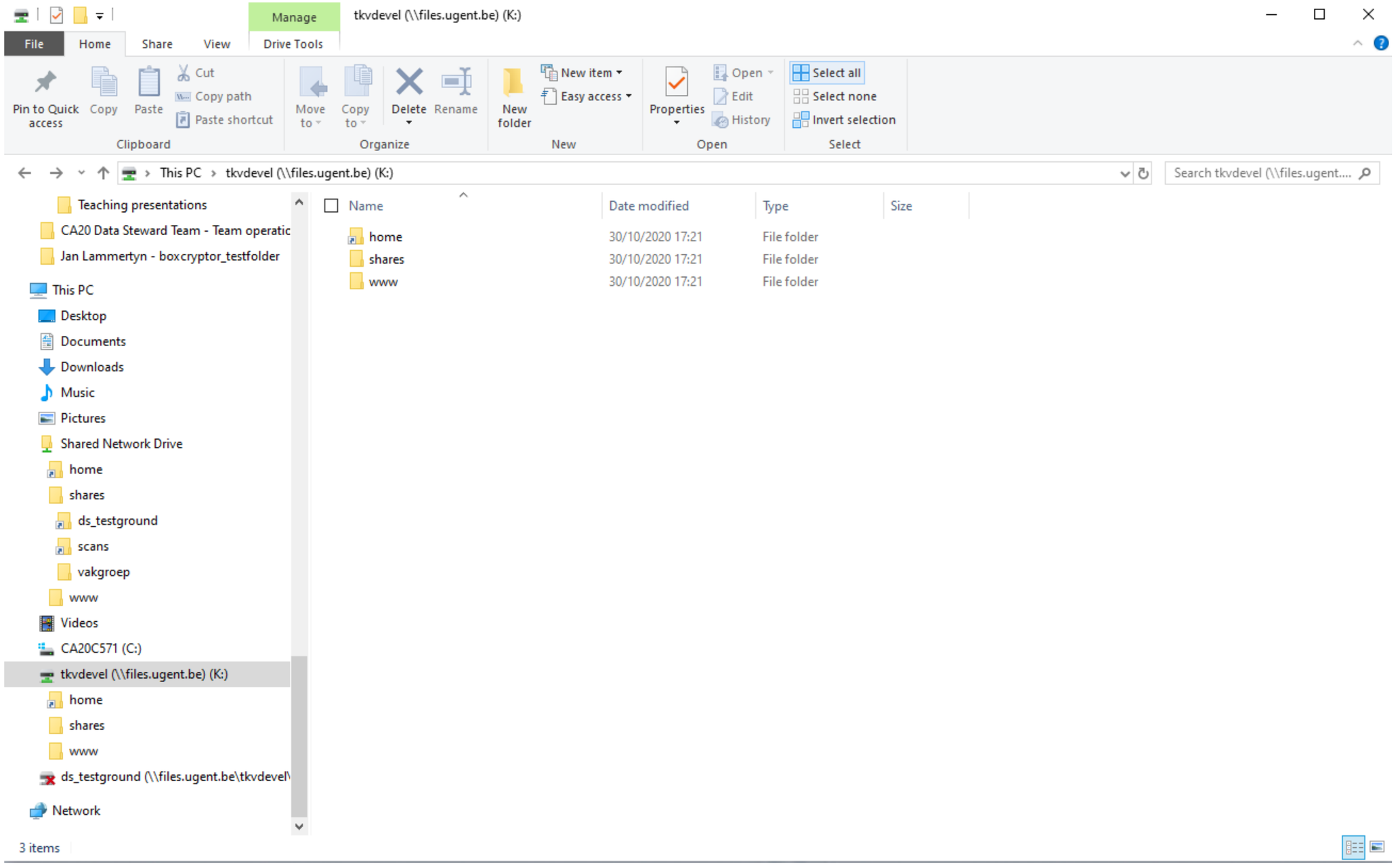
MOUNTING ON A LOCAL DEVICE



https://helpdesk.ugent.be/vpn/





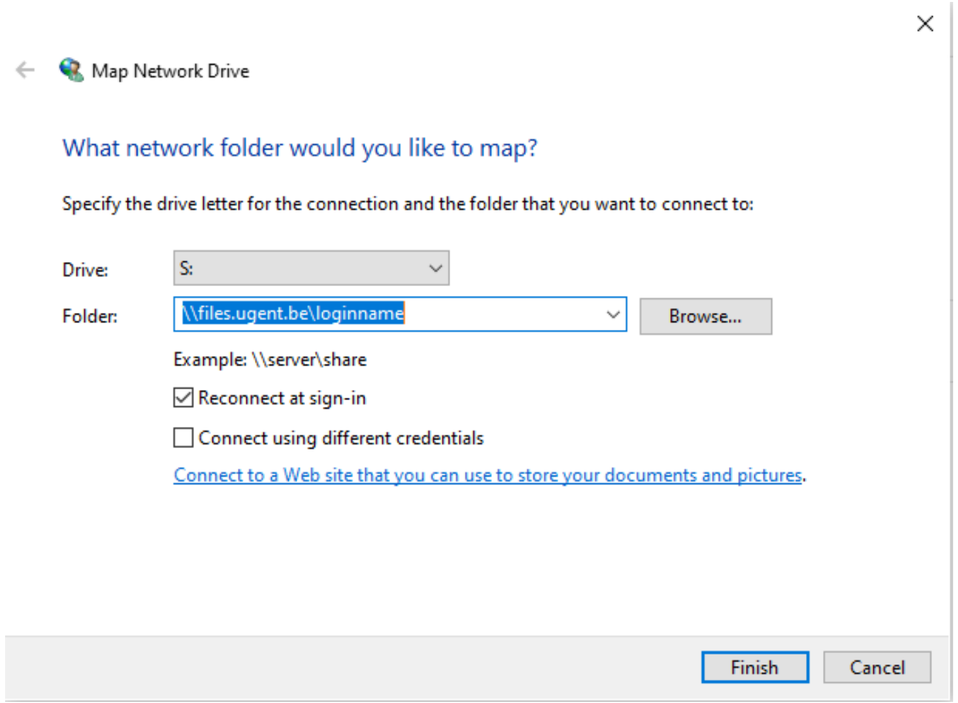


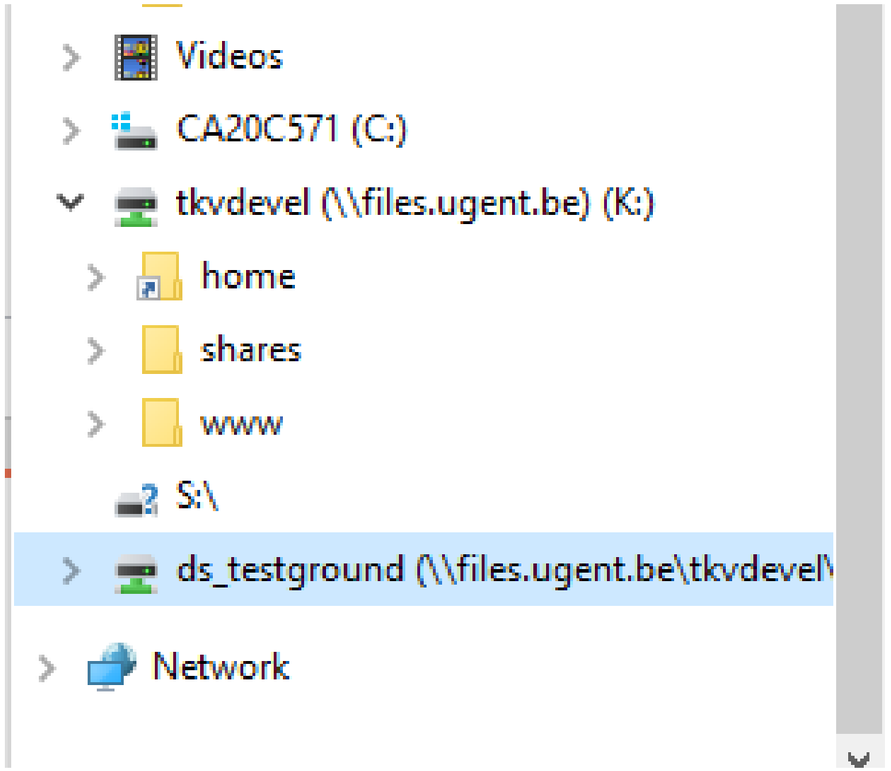
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MOUNTING A SPECIFIC SHARE

Mounting a specific share/location is also possible!

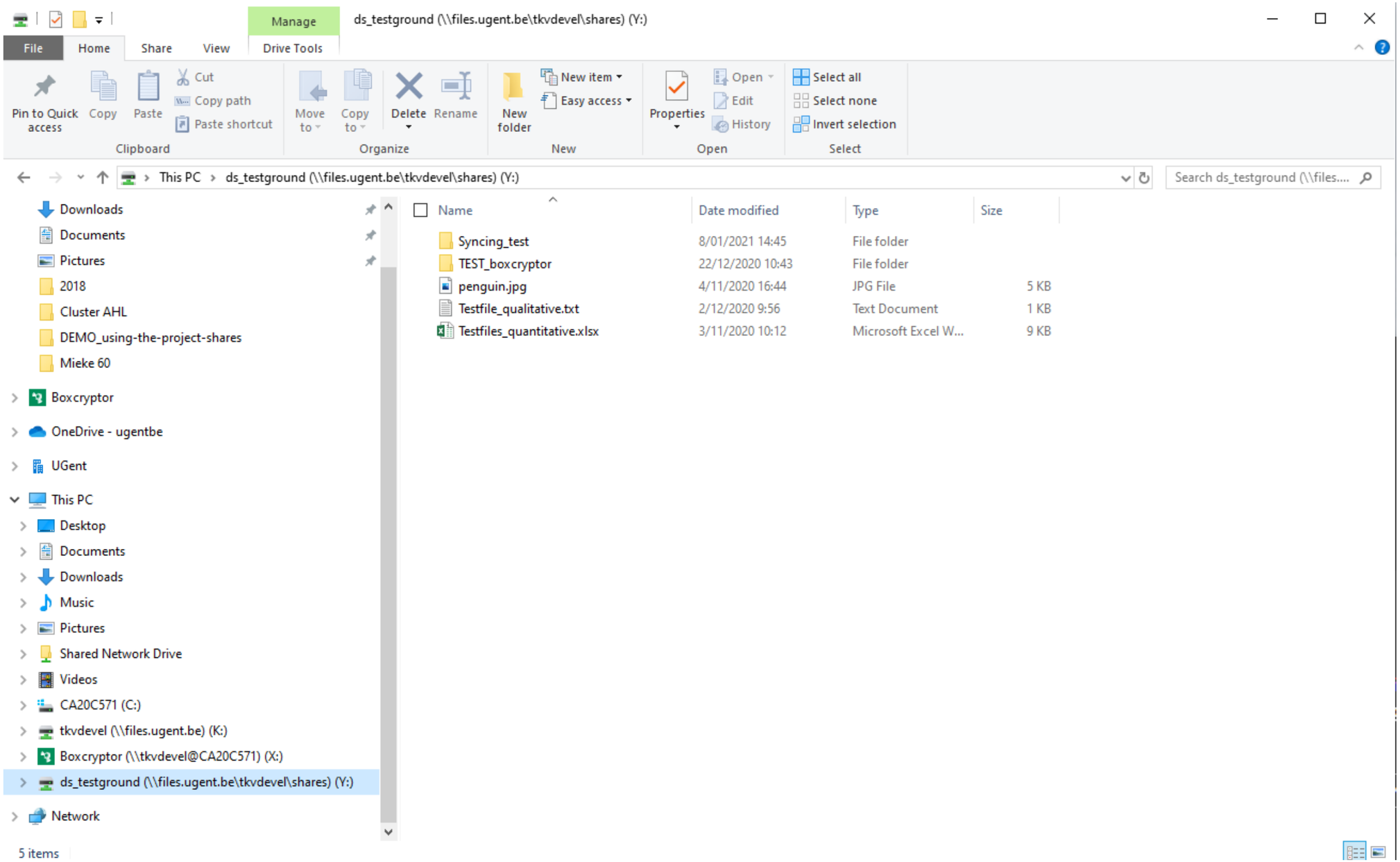
\\files.ugent.be\loginname\shares\name_of_the_share





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WORKING ON A SHARE

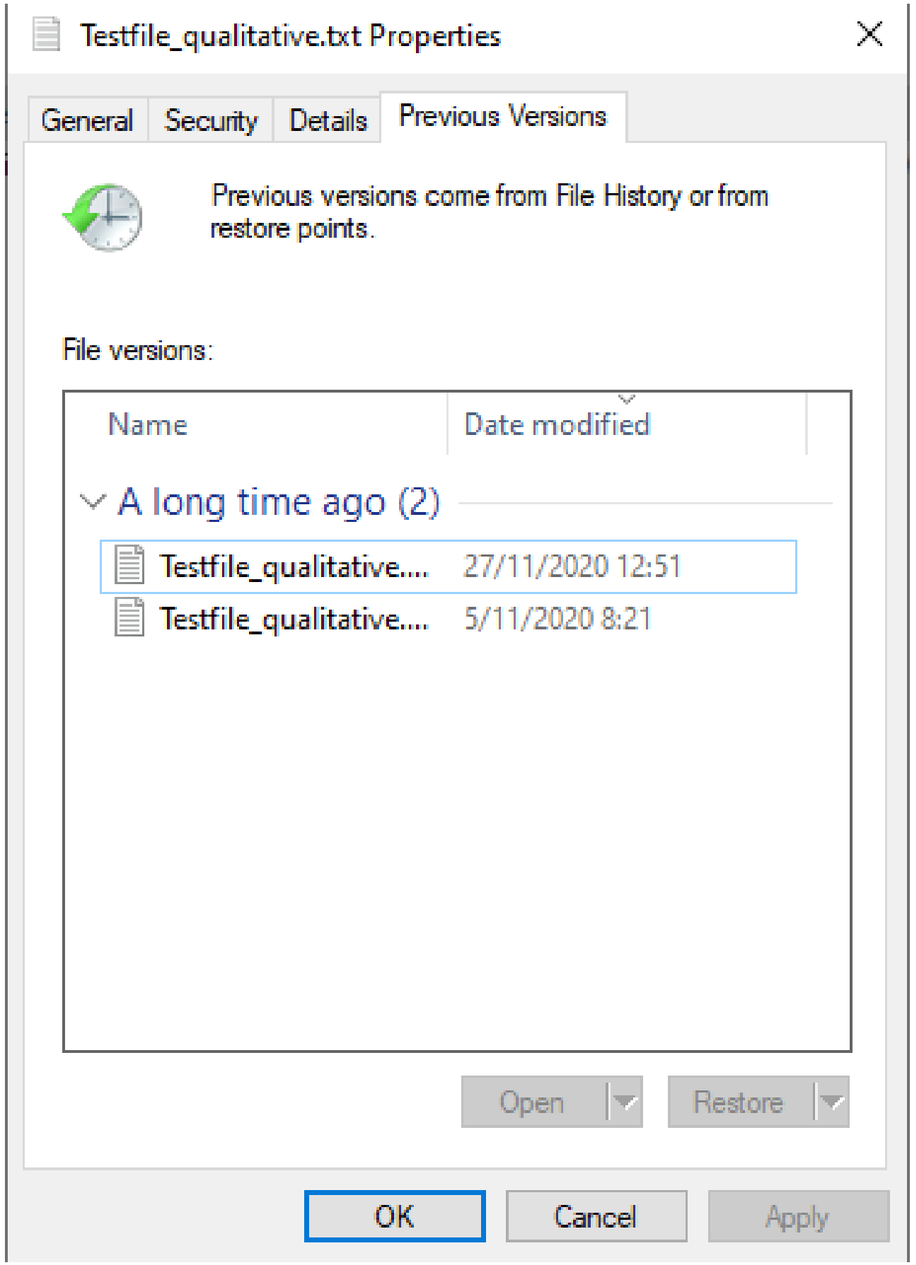


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THE PERKS OF SHARES: SNAPSHOTS

→5 weeks of daily snapshots

→15 weeks of weekly snapshots



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SHOWCASE:

- Requesting a share
- Mounting the share
- Using a share
- Using the snapshot feature

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PROJECT SHARES: 2 SCENARIOS

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GENERAL RULE



3-2-1 rule

- 3 copies
- 2 different types of media
- At least 1 backup on another location





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SCENARIO 1 – DIRECTLY ON THE PROJECT SHARE

+ Backups/snapshots are done automatically
+ Security is taken care of by DICT
+ No local copy, hence no security threat
+ Limited/controlled access
+ Archived/secured

– No off-line use
◦ VPN necessary outside UGent network

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ADDITIONAL BACKUP?: TO AN EXTERNAL HARD DISK

Additional layer of security?

- Incremental backup to an external device (e.g. external HD)

- Alternatively:
  - Working on OneDrive
  - Syncing to a Project Share
    - => 4 copies of your data….

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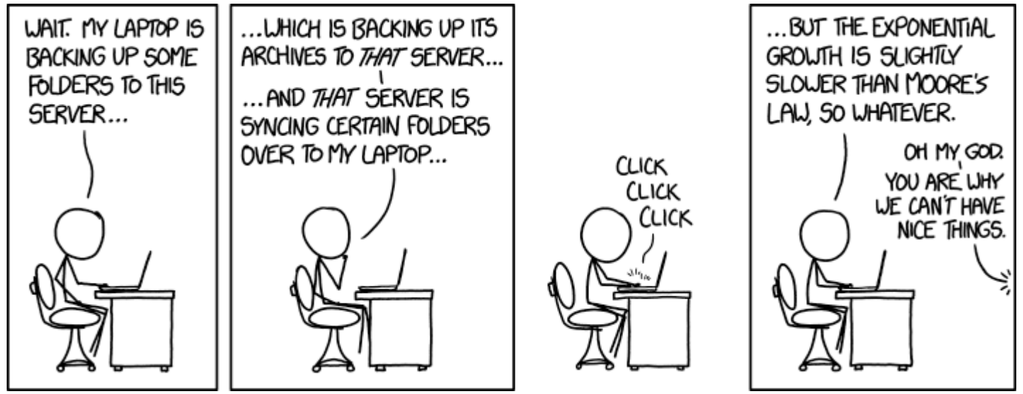
OPTION 2: SYNCHRONIZING FILES TO PROJECT SHARE

If you want to have backup copy of local files

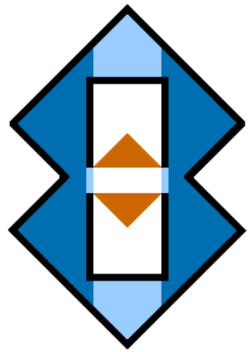
- Synchronize/mirror copy to project share
- Versioning/snapshots is done by DICT

Tools

- Different tools exist to synchronize files
- E.g., FreeFileSync, SyncBackFree



https://xkcd.com/1718/

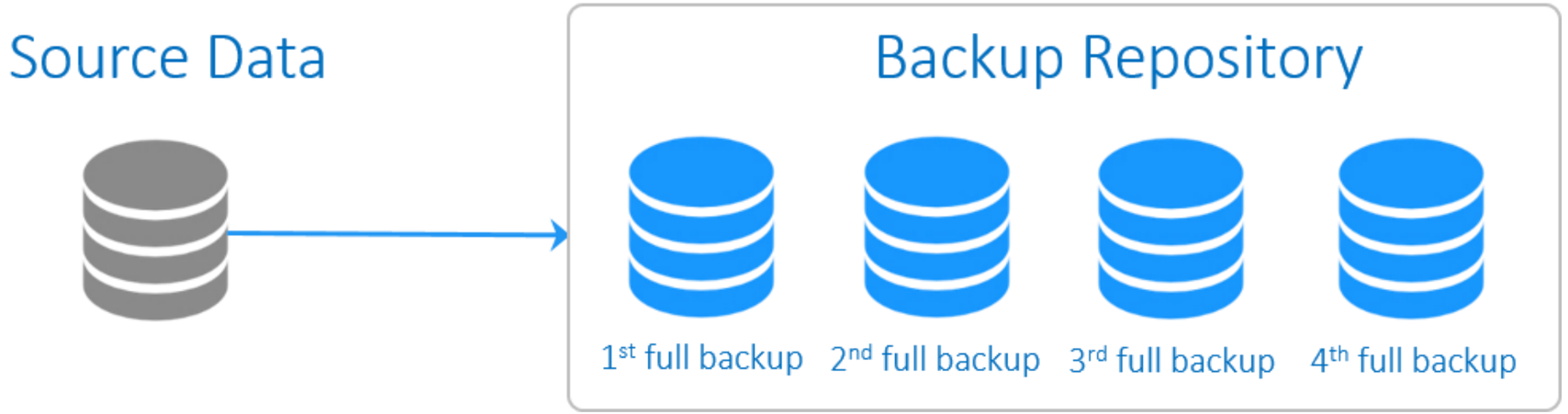




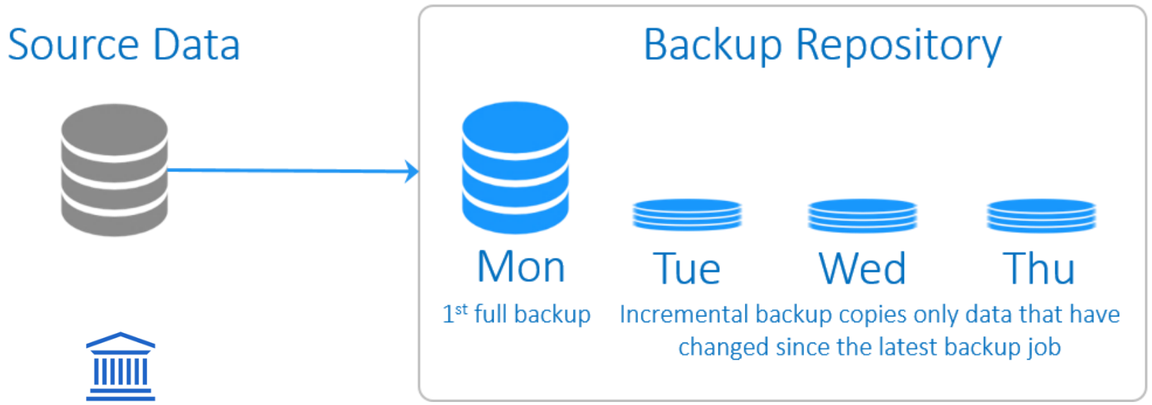
BACK-UP

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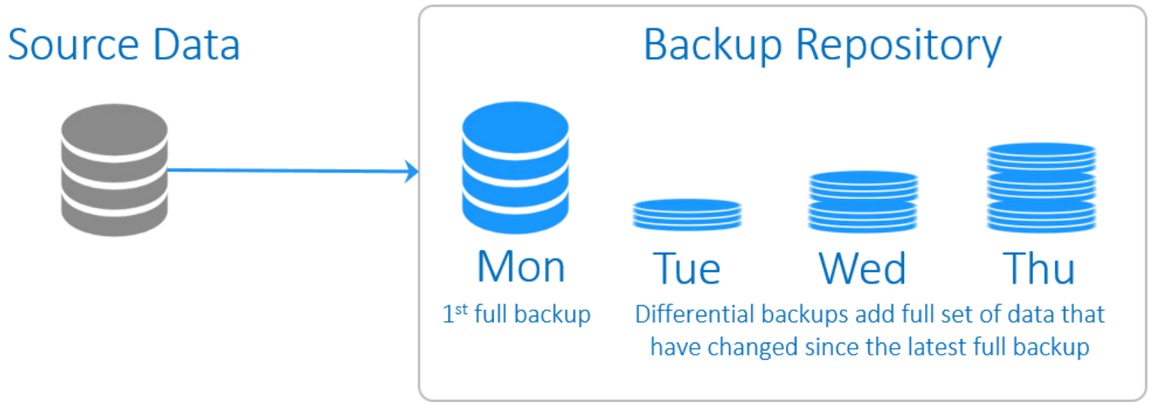
Full Backup



Incremental Backup



Differential Backup



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Images retrieved from https://www.nakivo.com/

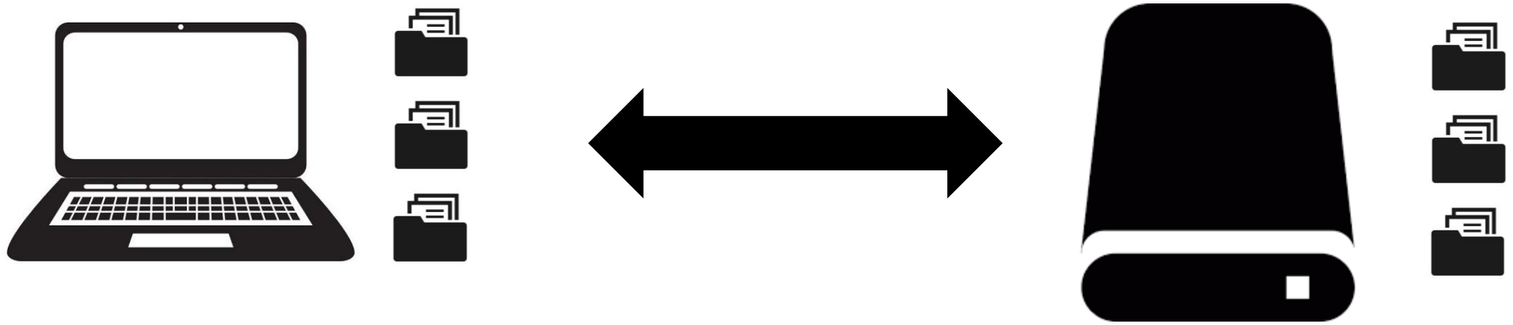
BACKUP, SYNCHRONIZATION, MIRRORING

BACKUP



SYNCHRONISATION

(two-way)



MIRRORING



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TEST YOUR BACKUP

Important!

- Put backups to the test.
- Check if and how you can retrieve files from your backup.
- Do not wait until you really need your backup.

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ugent.be/en/research/datamanagement

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